

Blended-Wing-Body Aerodynamics

CAD Design · SolidWorks CFD · 3D-Printed Model · Wind Tunnel Force-Balance Testing · $Re \approx 5.5 \times 10^4$

KEVIN ARMSTRONG II UC BERKELEY ME JAN - APR 2026 CFD + WIND TUNNEL



BWB 3D-printed model mounted on EWT three-component force-balance sting. 18x18 in test section, 5.0% blockage. Run 4: no-tuft configuration, AoA -2° to $+20^\circ$, $V_\infty \approx 40$ mph.

7.31

Peak L/D (CFD, 6° AoA)

14-16°

Stall onset (WT force data)

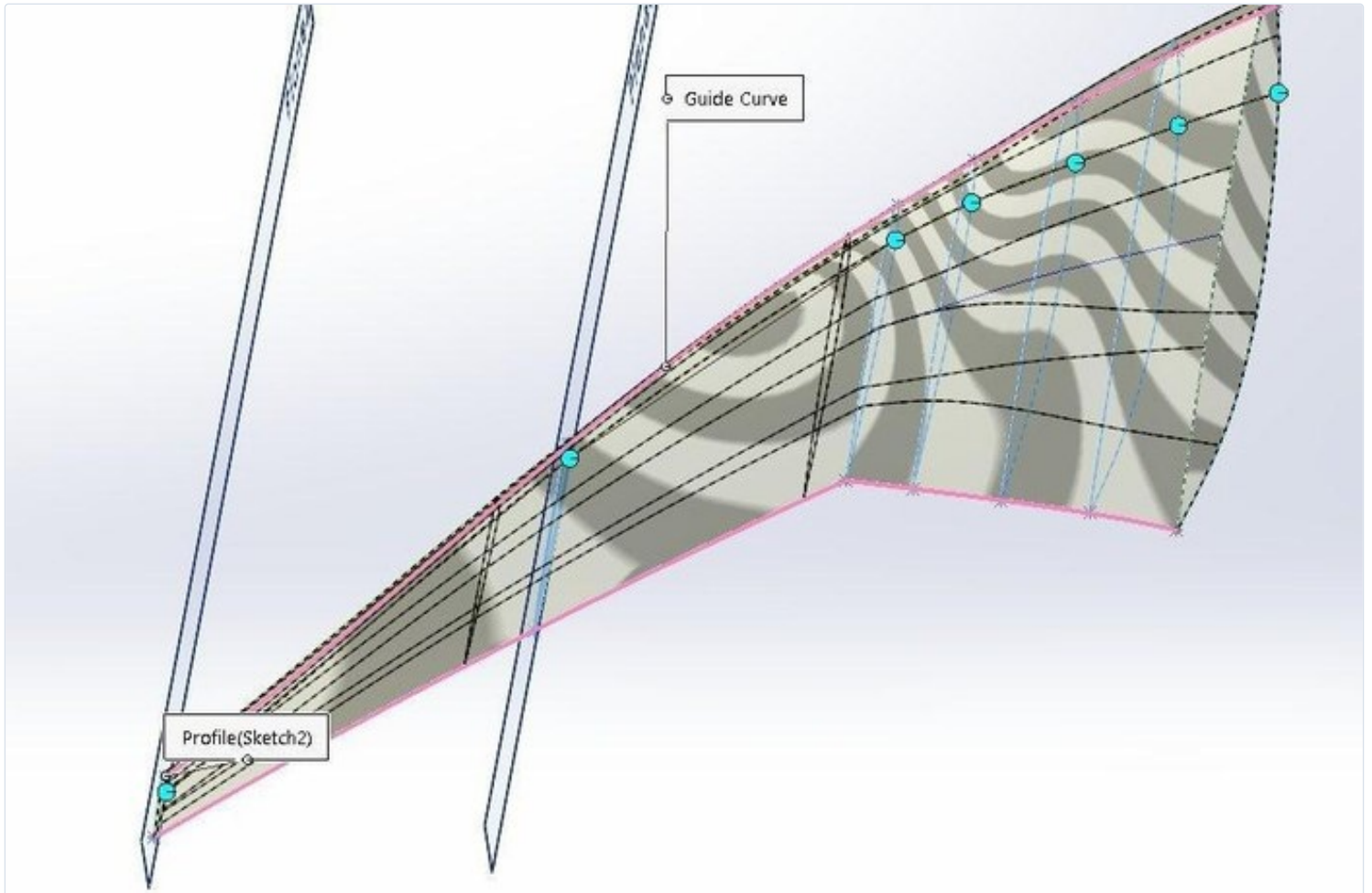
3.36 N

Peak normal force @ 18°

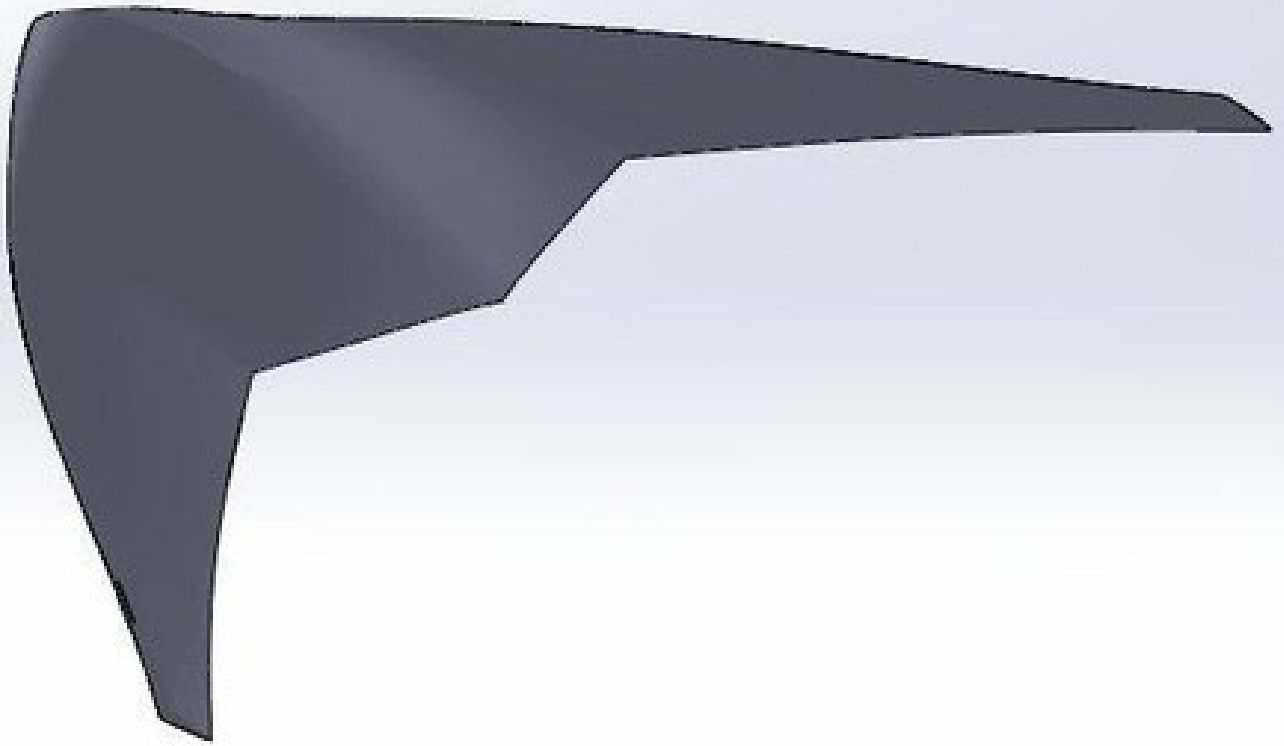
<0.5%

CFD domain sensitivity ΔCL

Geometry & Design



SOLIDWORKS multi-section loft with guide curves and cross-sectional profiles defining the BWB outer mold line.



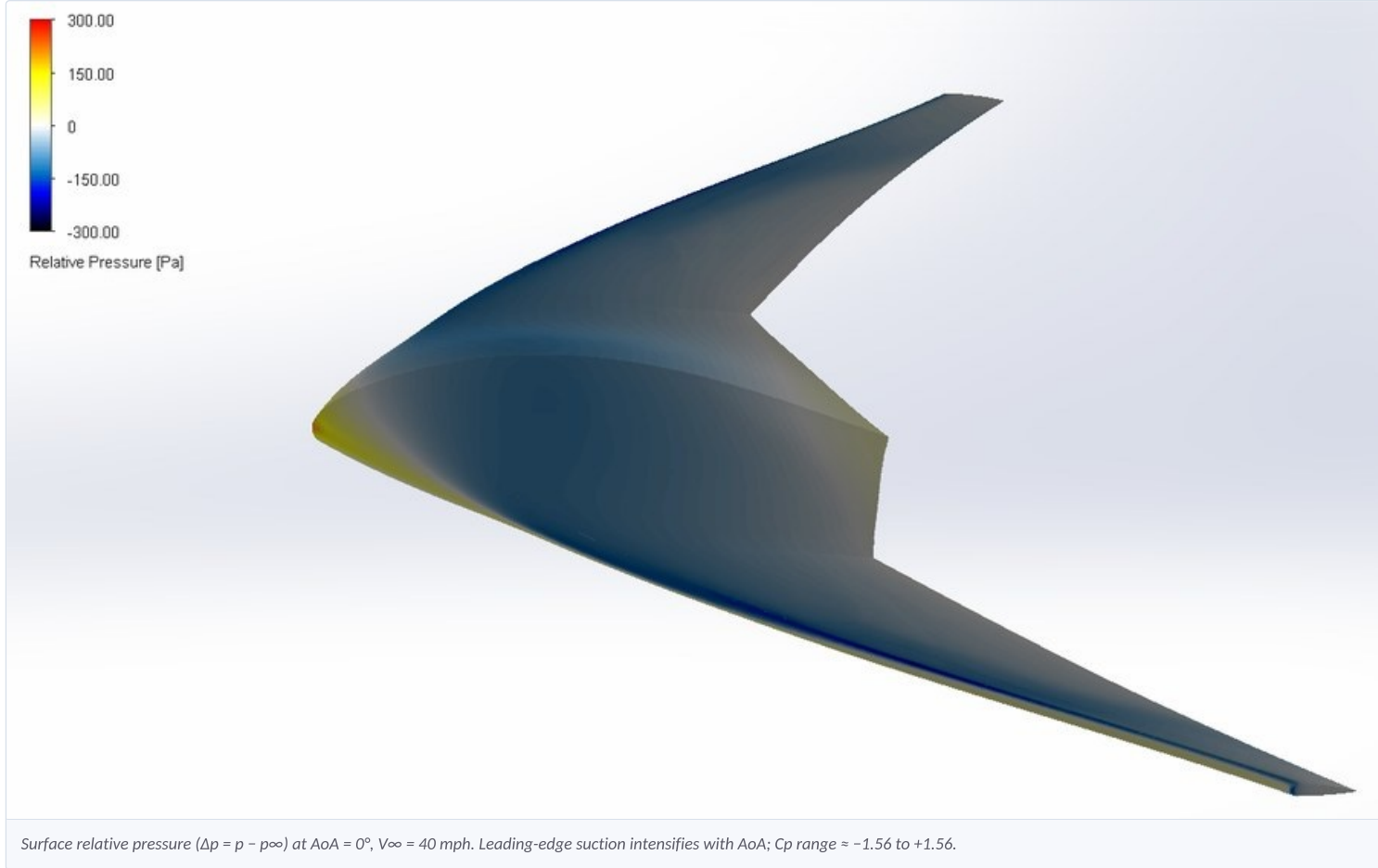
Rendered SOLIDWORKS model. Span 8 in (0.203 m), length 9 in (0.229 m), peak thickness 2 in.

The outer mold line was defined using a multi-section loft with four guide curves spanning centerbody to wingtip. The centerbody profile is a modified NACA 00-series section with an added volume term for blending and packaging, maintaining a smooth leading edge and tapered trailing edge. $S_{\text{ref}} = 9.62 \times 10^{-3} \text{ m}^2$ (top-view planform silhouette), $\bar{c} = 0.047 \text{ m}$.

SECTION 02

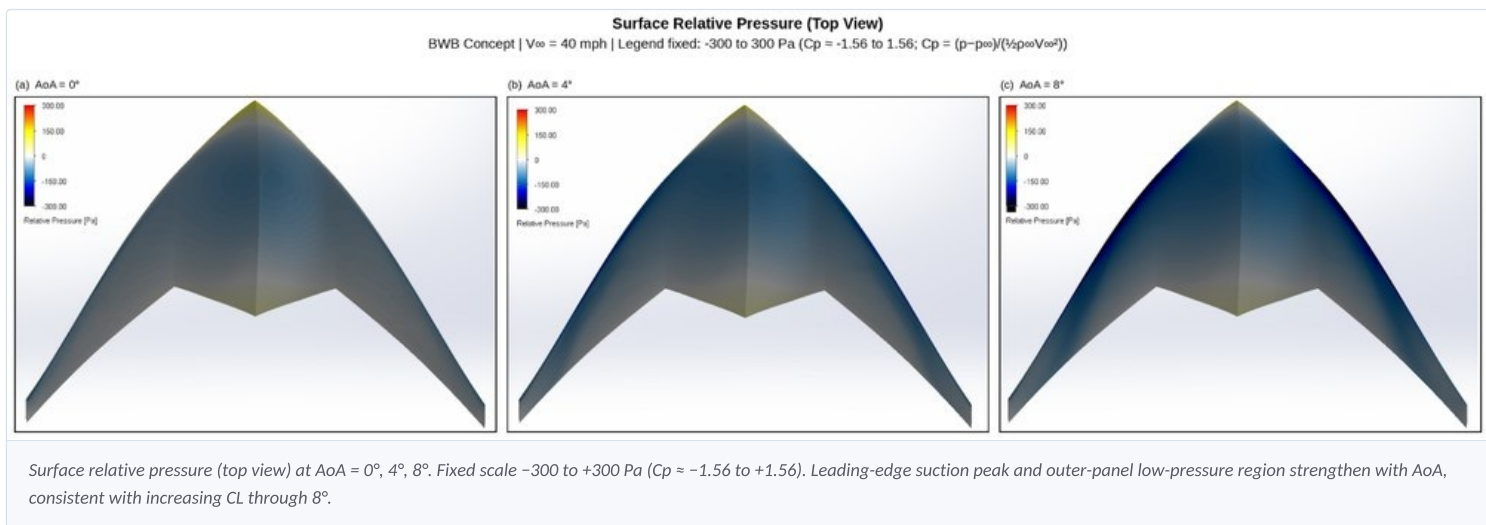
CFD: Validated Low-Speed Aerodynamics

SolidWorks Flow Simulation, steady-state, laminar + turbulent ($I = 1\%$, $L = 0.01 \text{ m}$), AoA swept from -2° to $+8^\circ$ at $V_\infty = 40 \text{ mph}$, $Re \approx 5.6 \times 10^4$. Verified with domain independence ($<0.5\% \Delta CL$, $<0.5\% \Delta CD$) and mesh refinement study (medium $\rightarrow$ fine $\Delta CL \approx 1.25\%$). Coefficients reported after +100-iteration post-convergence averaging window.



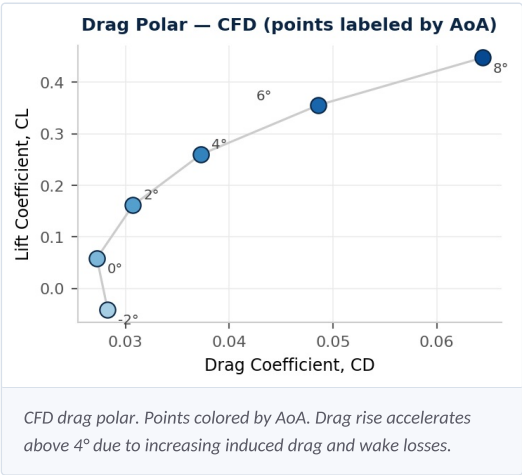
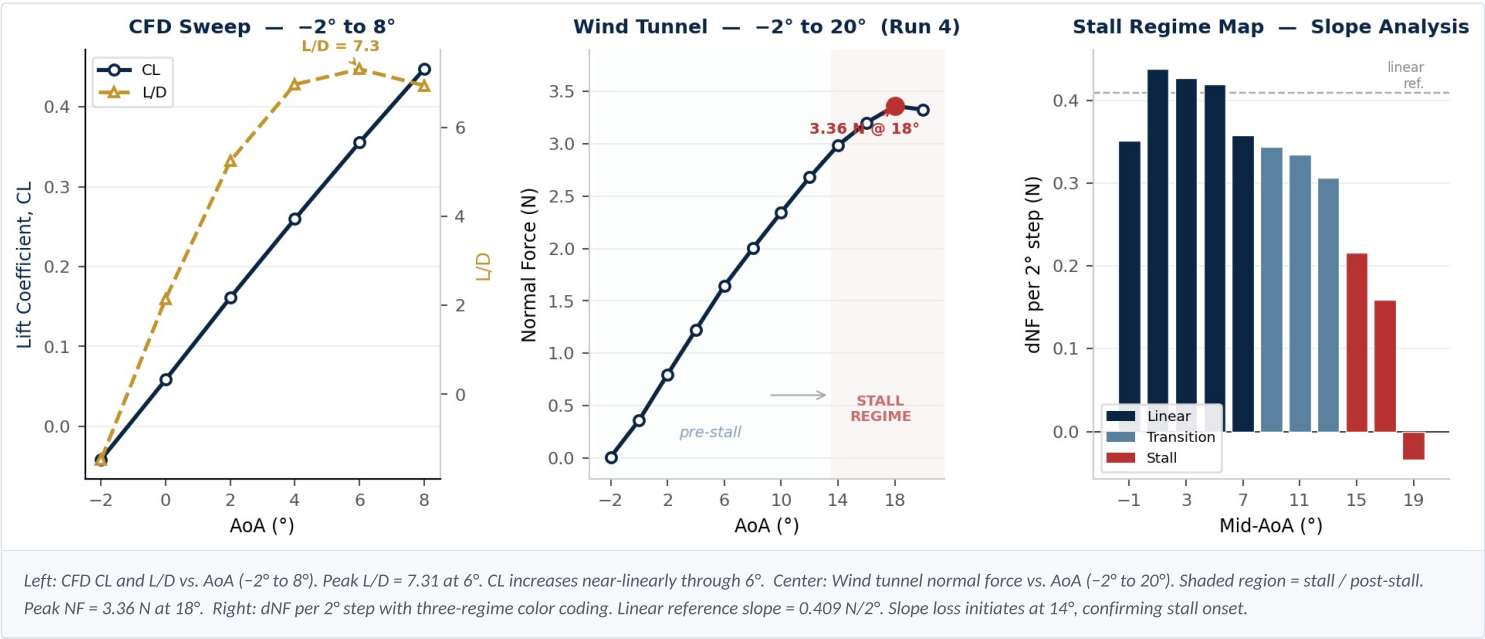
AoA (°)	CL	CD	L/D
-2	-0.0414	0.0283	-1.46
0	0.0586	0.0272	2.15
2	0.1612	0.0307	5.25
4	0.2598	0.0373	6.97
6	0.3554	0.0486	7.31
8	0.4473	0.0644	6.95

Mesh: large domain + medium cells (verified). $V_\infty = 40$ mph, $q_\infty = 192$ Pa. Bold = peak L/D.



Results: CFD Sweep + Wind Tunnel Stall Campaign

The CFD sweep characterizes pre-stall performance from -2° to 8° . Wind tunnel Run 4 (no-tuft, EWT balance, 25 Hz) extends the sweep to 20° AoA — capturing the full post-stall regime the simulation does not reach.



SECTION 04

Key Findings

FINDING 01

Stall is gradual — CFD did not predict it

The force-balance sweep extends 12° beyond the CFD range and captures a **progressive stall onset** at $14\text{--}16^{\circ}$ identified by two independent indicators: normal-force slope dropping below 50% of linear-regime reference *and* pitching-moment peak reversal at 16° . No sudden CL break — consistent with a blended, low-Re planform delaying abrupt leading-edge separation.

FINDING 02

CFD peak efficiency confirmed at 6° AoA

Verified L/D peak of 7.31 at 6° with numerically credible methodology: convergence + 100 stability iterations; domain sensitivity $<0.5\%$; mesh medium $\rightarrow$ fine Δ CL $\approx 1.25\%$. Results are repeatable and reported with consistent post-processing (fixed C_p scale, matched VX cut-planes) — not just decorative visuals.

FINDING 03

PM peak leads NF peak — AC shift near stall

Pitching moment peaks at 16° while normal force peaks at 18° . This 2° lead indicates the aerodynamic center is shifting forward near stall — a physical signature of outer-panel separation preceding the final lift peak. This detail is recoverable only with simultaneous PM measurement, not force-only testing.

Zero-lift angle $\approx -2^\circ$ — confirms cambered centerbody

Normal force at 0° AoA = **0.361 N** (non-zero). The zero-lift angle is approximately -1.5° to -2° , consistent with the NACA-derived cambered centerbody producing positive loading at zero incidence — the design intent. Provides a geometric cross-check on the CFD result where $CL(0^\circ) = +0.059$.

PORTFOLIO / RESUME BULLETS

Project Summary — Refined Bullets

PIPELINE Executed a full aerodynamics design-to-test pipeline on a custom blended-wing-body (BWB) — SOLIDWORKS multi-section loft, validated steady-state CFD (SolidWorks Flow Simulation), FDM fabrication, and EWT three-component force-balance wind tunnel testing — all at matched $Re \approx 5.5 \times 10^4$ with no scale correction required.

CFD Characterized pre-stall performance via validated CFD across -2° to $+8^\circ$ AoA (40 mph): peak $L/D \approx 7.3$ at 6° , CL increasing near-linearly; confirmed numerically with $<0.5\%$ domain sensitivity and medium→fine $\Delta CL \approx 1.25\%$; reported with consistent post-processing (fixed C_p scale, identical VX cut-planes across AoA).

WIND TUNNEL Extended the sweep to 20° AoA in wind tunnel testing, capturing a gradual stall the CFD range did not reach — identified onset at $14\text{--}16^\circ$ using dual indicators (normal-force slope collapse and pitching-moment reversal), measured peak NF of 3.36 N at 18° , and confirmed gradual separation behavior consistent with a low-Re blended planform.

METHODOLOGY Designed a rigorous multi-run balance campaign with wind-off tare procedure (WOT), tuft-on and tuft-free configurations, and point-by-point dynamic-pressure correction at 25 Hz / 25 samples per point — producing clean normal force, axial force, and pitching moment across a 12-point sweep with tunnel dynamic pressure stable within ± 1.5 Pa.